



Features of open data

There's just something so liberating about being able to find information on any subject at all on the Internet. Can't remember the 3rd line of your favorite childhood song, curious who had the most home runs in 1986, want to teach yourself sign language? Just pop open your laptop, type up some text and poof, you have what you need. Many groups think we should also have this level of access to data. There's even a global movement that believes the openness of data can transform society and how decisions are made. So far, we've talked a lot about the power of data and the importance of data ethics concerns including ownership, transaction transparency, consent, currency, and privacy. Now, let's talk about openness. When referring to data, **openness refers to free access, usage and sharing of data**. Sometimes we **refer to this as open data**, but it doesn't mean we ignore the other aspects of data ethics we covered. We should still be transparent, respect privacy, and make sure we have consent for data that's owned by others. This just means we can access, use, and share that data if it meets these high standards. For example, there are standards around availability and access. Open data must be available as a whole, preferably by downloading over the Internet in a **convenient** and **modifiable** form. The website data.gov is a great example. You can download science and research data for a wide range of industries in simple file formats like a spreadsheet. Another standard surrounds **reuse and redistribution**. Open data

must be provided under terms that allow reuse and redistribution including the ability to use it with other datasets. And the last area is universal participation. Everyone must be able to use, reuse, and redistribute the data. There shouldn't be any **discrimination** against fields, persons, or groups. No one can place restrictions on the data like making it only available for use in a specific industry. Now let's talk a little more about why open data is such a great thing and how it can help you as a data analyst. **One of the biggest benefits of open data is that credible databases can be used more widely.** More importantly, all of that good data can be **leveraged**, shared, and combined with other data. **Just imagine the impact that would have on scientific collaboration, research advances, analytical capacity, and decision-making.** For example, in human health, openness allows us to access and combine diverse data to detect diseases earlier and earlier. In government, you can help hold leaders **accountable** and provide better access to community services. The possibilities and benefits are almost endless. But of course, every big idea has its challenges. **A whole lot of resources are needed to make the technological shift to open data.** **Interoperability** is key to open data's success. **Interoperability is the ability of data systems and services to openly connect and share data.** For example, data interoperability is important for health care information systems where multiple organizations such as hospitals, clinics, pharmacies, and laboratories need to access and share data to ensure patients get the care that they need. This is why your doctor is able to send your prescription directly to your pharmacy to fill. They have **compatible** databases that allow them to share information. But this kind of interoperability requires a lot of cooperation. While there is serious potential in the open, timely, fair, and simple sharing of data, its future will depend on how effectively larger

challenges are addressed. As a data analyst, I say **the sooner the better**. Speaking of which, we're going to talk more about open data and see its use in action in an upcoming video. Now that you've learned all about data ethics, you have some important principles to guide you on your data journey. Anytime you're not sure of your data, remember what you've learned here. Happy Trails.